

MH1301 Discrete Mathematics

Handout 6: Linear Recurrence Relations (2)

This is the last handout for Part 2 on Recurrence Relations. We focus on solving recurrence relations that are linear and non-homogeneous.

- Linear non-homogeneous recurrence relations.

Linear Non-Homogeneous Recurrence

A linear non-homogeneous recurrence relation of
is a recurrence relation of the form

$$a_n = c_1 a_{n-1} + c_2 a_{n-2} + \cdots + c_k a_{n-k} + F(n)$$

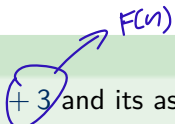
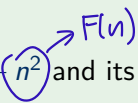
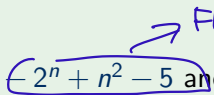
where $c_1, \dots, c_k$ are real numbers and $c_k \neq 0$ and $F(n) \neq 0$ is a function depending only on n .

- associated homogeneous recurrence relation, i.e.,

$$b_n = c_1 b_{n-1} + c_2 b_{n-2} + \cdots + c_k b_{n-k}.$$

Linear Non-Homogeneous Recurrence

Examples

- $f_n = f_{n-1} + f_{n-2} + 3$ and its associated homogeneous recurrence is $b_n = b_{n-1} + b_{n-2}$.

- $a_n = 3a_{n-1} + 5a_{n-2} + n^2$ and its associated homogeneous recurrence is $b_n = 3b_{n-1} + 5b_{n-2}$.

- $a_n = 2a_{n-1} - 5a_{n-3} - 2^n + n^2 - 5$ and its associated homogeneous recurrence is $b_n = 2b_{n-1} - 5b_{n-3}$.


Linear Non-Homogeneous Recurrence

Question

Forget about initial conditions for now. How many solutions can you find?
for the recurrence relation

$$a_n = c_1 a_{n-1} + c_2 a_{n-2} + \cdots + c_k a_{n-k} + F(n)?$$

If (a_n) and (b_n) are two solutions, is $(a_n + b_n)$ also a solution?

$$a_n = c_1 a_{n-1} + \cdots + c_k a_{n-k} + F(n),$$

$$b_n = c_1 b_{n-1} + \cdots + c_k b_{n-k} + F(n),$$

then

$$(a_n + b_n) = c_1(a_{n-1} + b_{n-1}) + \cdots + c_k(a_{n-k} + b_{n-k}) + \underbrace{2 F(n)}_{\substack{\downarrow \\ \text{not what} \\ \text{we want!}}}.$$

So $(a_n + b_n)$ is not a solution!

Linear Non-Homogeneous Recurrence

Question

Forget about initial conditions for now. How many solutions can you find?
for the recurrence relation

$$a_n = c_1 a_{n-1} + c_2 a_{n-2} + \cdots + c_k a_{n-k} + F(n)?$$

However...

not solⁿ for the whole thing!

If (a_n) is a solution and (b_n) is a solution to the associated homogeneous recurrence relation, then

$$a_n = c_1 a_{n-1} + \cdots + c_k a_{n-k} + F(n),$$

$$b_n = c_1 b_{n-1} + \cdots + c_k b_{n-k},$$

$$\text{and } (a_n + b_n) = c_1(a_{n-1} + b_{n-1}) + \cdots + c_k(a_{n-k} + b_{n-k}) + F(n).$$

Thus $(a_n + b_n)$ is a solution now!

Linear Non-Homogeneous Recurrence

Theorem 5 (Section 8.2)

Given recurrence relation

$$a_n = c_1 a_{n-1} + c_2 a_{n-2} + \cdots + c_k a_{n-k} + F(n),$$

if $(a_n^{(p)})$ is a particular solution $\rightarrow$ solⁿ of the whole thing of the recurrence, then every solution of the recurrence is of the form

$$(a_n^{(p)} + b_n)$$

$\rightarrow$ solⁿ only for the associated part

where (b_n) is the solution of its associated homogeneous recurrence

$$b_n = c_1 b_{n-1} + c_2 b_{n-2} + \cdots + c_k b_{n-k}.$$

Note: b_n has NO initial condition!

Proof. Since $(a_n^{(p)})$ is a particular solution of the nonhomogeneous recurrence relation, we know that

$$a_n^{(p)} = c_1 a_{n-1}^{(p)} + c_2 a_{n-2}^{(p)} + \cdots + c_k a_{n-k}^{(p)} + F(n). \quad (1)$$

Suppose (d_n) is another solution of the nonhomogeneous recurrence relation, i.e.,

$$d_n = c_1 d_{n-1} + c_2 d_{n-2} + \cdots + c_k d_{n-k} + F(n) \quad (2)$$

Then subtracting (1) from (2) gives

$$\begin{aligned} d_n - a_n^{(p)} &= c_1 (d_{n-1} - a_{n-1}^{(p)}) + c_2 (d_{n-2} - a_{n-2}^{(p)}) + \cdots + c_k (d_{n-k} - a_{n-k}^{(p)}) + 0. \end{aligned}$$

It follows that $(d_n - a_n^{(p)})$ is a solution of the associated homogeneous

linear recurrence. It remains to set $b_n = d_n - a_n^{(p)}$ for all n . (i.e. $d_n = \underbrace{a_n^{(p)}}_{\text{seen on slide 7}} + b_n$)

Example 10 (Section 8.2)

Find all solutions of the relation $a_n = 3a_{n-1} + 2n$ with $a_1 = 3$.

← want $a_n^{(p)}$ to satisfy this

Solution:

$r=3$

- Its associated homogeneous recurrence is $b_n = 3b_{n-1}$, whose solution is

$$b_n = 3 \cdot b_{n-1} = 3^2 \cdot b_{n-2} = 3^3 \cdot b_{n-3} = \dots = 3^n \cdot b_0 = h \cdot 3^n,$$

where h is a constant.

- We need to find a particular solution. $F(n) = 2n$ is a polynomial in n of degree 1, let's try a linear function in n , say $c \cdot n + d$, where c and d are constants.
- If $a_n^{(p)} = c \cdot n + d$ is a solution, then the recurrence becomes

$$c \cdot n + d = 3(c \cdot (n-1) + d) + 2n.$$

$$a_n = -n - 1.5$$

$$\therefore (2 + 2c)n + (2d - 3c) = 0$$

$$\therefore 2 + 2c = 0 \quad \text{and} \quad 2d - 3c = 0$$

$$\therefore c = -1 \quad \text{and} \quad d = -1.5.$$

Example 10 (Section 8.2)

- Hence, $a_n = -n - 1.5$ is a particular solution.
- By Theorem 5, all solutions of the recurrence $a_n = 3a_{n-1} + 2n$ are of the form

$$a_n = a_n^{(p)} + b_n = \underbrace{-n - 1.5}_{a_n^{(p)}} + \underbrace{h \cdot 3^n}_{b_n}$$

where h is a constant.

- By initial condition $a_1 = 3$, we have

$$-1 - 1.5 + h \cdot 3 = 3 \Rightarrow h = 11/6.$$

- Hence, the solution of the recurrence is

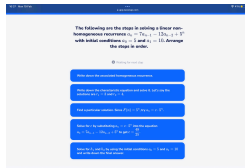
$$a_n = -n - 1.5 + (11/6)3^n.$$

Check

$$a_1 = -1 - 1.5 + \frac{11}{6}(3) = 3 \Rightarrow \text{correct}$$

$$a_2 = -2 - 1.5 + \frac{11}{6}(3)^2 = 13$$

$$a_3 = -3 - 1.5 + \frac{11}{6}(3)^3 = 45$$



Example

Find all solutions of the relation $a_n = -2a_{n-1} + \underbrace{n+1}_{F(n)}$ with $a_0 = 2$.

Solution:

- Its associated homogeneous recurrence is $b_n = -2b_{n-1}$, whose solution is $b_n = (-2) \cdot b_{n-1} = (-2)^2 \cdot b_{n-2} = (-2)^3 \cdot b_{n-3} = \dots = (-2)^n \cdot b_0 = \underbrace{h \cdot (-2)^n}_{\text{based on } F(n)}$, where h is a constant.
- We need to find a particular solution. Let's still guess $\underbrace{c \cdot n + d}_{\text{degree 1 polynomial}}$, where c and d are constants.
- If $a_n = c \cdot n + d$ is a solution, then the recurrence becomes

$$\begin{aligned}c \cdot n + d &= -2(c \cdot (n-1) + d) + n + 1 \\ \Rightarrow (1 - 3c)n + (1 + 2c - 3d) &= 0 \\ \Rightarrow 1 - 3c = 0 \quad \text{and} \quad 1 + 2c - 3d &= 0 \\ \Rightarrow c = 1/3 \quad \text{and} \quad d = 5/9.\end{aligned}$$

Example

- Hence, $a_n^{(p)} = n/3 + 5/9$ is a particular solution. ↗ particular solⁿ

- By Theorem 5, all solutions of the recurrence $a_n = 3a_{n-1} + 2n$ are of the form

$$a_n = a_n^{(p)} + b_n = n/3 + 5/9 + h \cdot (-2)^n$$

where h is a constant. $a_0 = 0/3 + 5/9 + h \cdot (-2)^0$

- By initial condition $a_0 = 2$, we have

$$2 = 5/9 + h \Rightarrow h = \frac{13}{9}.$$

- Hence, the solution of the recurrence is

$$a_n = \frac{n}{3} + \frac{5}{9} + \left(\frac{13}{9}\right)(-2)^n.$$

Example 11 (Section 8.2)

degree 2 recurrence

Find all solutions of the relation $a_n = 5a_{n-1} - 6a_{n-2} + 7^n$.

Solution:

$F(n) \rightarrow$ power function

- Its associated homogeneous recurrence is $b_n = 5b_{n-1} - 6b_{n-2}$.
- Its characteristic equation is $r^2 - 5r + 6 = 0$, whose roots are $r_1 = 2$ and $r_2 = 3$. Hence, its solution is $b_n = h_1 2^n + h_2 3^n$.
- We need to find a particular solution. Since $F(n) = 7^n$, let's try $c \cdot 7^n$, where c is a constant.
- If $a_n = c \cdot 7^n$ is a solution, then the recurrence becomes

$$\begin{aligned} c \cdot 7^n &= 5c \cdot 7^{n-1} - 6c \cdot 7^{n-2} + 7^n \\ \text{divide by } 7^{n-2} \quad \therefore \quad 49c &= 35c - 6c + 49 \\ \therefore \quad c &= 49/20. \end{aligned}$$

Example 11 (Section 8.2)

- Hence, $a_n = (49/20)7^n$ is a particular solution.
- By Theorem 5, all solutions of the recurrence $a_n = 5a_{n-1} - 6a_{n-2} + 7^n$ are of the form

$$a_n = a_n^{(p)} + b_n = (49/20) \cdot 7^n + h_1 \cdot 2^n + h_2 \cdot 3^n$$

where h_1 and h_2 are constants.

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Question

How to find a particular solution to a linear nonhomogeneous recurrence relation?

Find a Particular Solution

Suppose that (a_n) satisfies the linear nonhomogeneous recurrence relation

$$a_n = c_1 a_{n-1} + c_2 a_{n-2} + \cdots + c_k a_{n-k} + F(n)$$

where $F(n) = (b_t n^t + b_{t-1} n^{t-1} + \cdots + b_1 n + b_0)$ is a **polynomial**.

- Try $a_n^{(p)} = p_t n^t + p_{t-1} n^{t-1} + \cdots + p_1 n + p_0$.
when **1 is not** a root of the characteristic equation of the associated homogeneous recurrence relation.
- Try $a_n^{(p)} = n \cdot (p_t n^t + p_{t-1} n^{t-1} + \cdots + p_1 n + p_0)$.
when **1 is** a root of the characteristic equation of the associated homogeneous recurrence relation.

$$\hookrightarrow h_1 \cdot 1^n + h_2 \cdot r_2^n + \dots = h_1 + h_2 r$$

Find a Particular Solution

Suppose that (a_n) satisfies the linear nonhomogeneous recurrence relation

$$a_n = c_1 a_{n-1} + c_2 a_{n-2} + \cdots + c_k a_{n-k} + F(n)$$

where $F(n) = s^n$ is an **exponential**.

- Try $a_n^{(p)} = c \cdot s^n$.
when s **is not** a root of the characteristic equation of the associated homogeneous recurrence relation.
- Try $a_n^{(p)} = c \cdot n \cdot s^n$.
when s **is** a root of the characteristic equation of the associated homogeneous recurrence relation.

Recurrence Relation - Review

Recurrence Relation

- For a sequence (a_n) , a recurrence relation is a formula that relates each term a_n to certain of its predecessors $a_{n-1}, a_{n-2}, \dots, a_{n-k}$, where $k \leq n$ is an integer. It involves
 - ▶ recurrence formula and
 - ▶ initial values.
- Find the connection between problem of size n with those with size $n-1, n-2, \dots$

Recurrence Relation - Review

When solving Recurrence Relation, use

- **Theorem 1.** For linear homogeneous recurrence with degree 2 and two distinct characteristic roots.
- **Theorem 2.** For linear homogeneous recurrence with degree 2 and one distinct characteristic roots.
- **Theorem 3.** For linear homogeneous recurrence with degree k and k distinct characteristic roots.
- **Theorem 5.** For linear non-homogeneous recurrence.

Reading Assignment

Rosen (7th edition):

- Chapter 8.1, Applications of Recurrence Relations
- Chapter 8.2, Solving Linear Recurrence Relations